Definition, meaning, and measurement of the polarization extinction ratio of fiber-based devices

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We clarify the definition of the polarization extinction ratio—also called polarization cross talk—of fiber-based devices. Its strong wavelength dependence, even for simple devices such as a single-fiber patchcord, is highlighted. Thus white sources may not be used for most measurements. We also explain the weakness of measurements with a rotating polarizer and a monochromatic source. Only a polarimeter may be used if accuracy is required. We report measurements of connections (including two connectors and one mating sleeve) and show the importance of the mating sleeve in the result. Finally, we define the validity domain of the standardized method, which uses both a white source and a rotating polarizer. © 2005 Optical Society of America

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1. Introduction

The polarization extinction ratio (PER), also called polarization cross talk, in polarization-maintaining (PM) fiber-based devices is a key parameter.[1–4] It is also an important parameter of polarizing devices.[5,6] It is defined as the ratio between the optical power of the signal on the state of polarization (SOP) on which it is supposed to propagate P_x and the optical power on the orthogonal axis P_y. As a matter of fact, when the signal is launched on one SOP of a PM device, it is supposed to remain on it. Thus the PER seems to be a good figure of merit. However, we show that, even for a very simple device such as one single-fiber patchcord composed of one PM fiber (PMF) and two connectors, the PER is wavelength[7] (i.e., frequency) and time[8] dependent. Hence the signal suffers not only from loss, which can easily be compensated for with an optical amplifier, but also from distortions when its spectrum bandwidth is of the order of (or larger than) the frequency scale of the PER. This effect is not anecdotal because the frequency scale is determined by the group birefringence of the device. For instance, a PMF exhibits differential group delay (DGD) between its principal axes of the order of 1 or 2 ps/m, depending on the manufacturing process and on the wavelength. Hence, with one PMF of a dozen meters or a few concatenated PMFs, telecom 40-Gbit/s signals may suffer from distortions. This is also an important problem for high-power lasers such as those of the Laser MegaJoule (LMJ) or the National Ignition Facility (NIF).[9] As a matter of fact, in that case, the pulses are phase modulated to avoid stimulated Brillouin scattering and to smooth the beam.[7] The signal bandwidth is of the order of a few tenths of gigahertz. The distortions consist here of the conversion of part of the phase modulation into intensity modulation and are known under the name[7] "FM–AM conversion."

Therefore an accurate measurement of the PER is critical. It cannot be measured with a white source as this would result in a kind of averaging over wavelength. Moreover, the measured quantity would not even be the mean PER over wavelength. As a matter of fact, the former is equal to the ratio of the intensities averaged over wavelength $\langle P_x\rangle/\langle P_y\rangle$, whereas the latter is the averaged ratio over wavelength $\langle P_x/P_y\rangle$. As opposed to Refs. 10 and 11, here an intensity formalism is not relevant. As an example, consider the simple case of two rotations of angles θ_1 and θ_2 at both sides of one birefringent element, typically modeling a PM patchcord. With an intensity formalism, the equation would simply be[10,11]

$$\begin{pmatrix} P_x \\ P_y \end{pmatrix} = \begin{bmatrix} \cos^2\theta_2 & \sin^2\theta_2 \\ \sin^2\theta_2 & \cos^2\theta_2 \end{bmatrix} \begin{bmatrix} \cos^2\theta_1 & \sin^2\theta_1 \\ \sin^2\theta_1 & \cos^2\theta_1 \end{bmatrix} \begin{pmatrix} P_{x0} \\ 0 \end{pmatrix}, \quad (1)$$

and PER seems to be equal to $\mathrm{PER}_{\mathrm{Int}}$, defined by

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$$\mathrm{PER}_{\mathrm{Int}} = \frac{\cos^2\theta_1\cos^2\theta_2 + \sin^2\theta_1\sin^2\theta_2}{\cos^2\theta_1\sin^2\theta_2 + \sin^2\theta_1\cos^2\theta_2}. \quad (2)$$

However, the relevant formalism is based on fields, and the equation becomes

$$\begin{bmatrix}\sqrt{P_x}\exp(i\varphi_x)\\ \sqrt{P_y}\exp(i\varphi_y)\end{bmatrix} = \begin{bmatrix}\cos\theta_2 & -\sin\theta_2\\ \sin\theta_2 & \cos\theta_2\end{bmatrix} \times \begin{bmatrix}\exp(i\varphi/2) & 0\\ 0 & \exp(-i\varphi/2)\end{bmatrix} \times \begin{bmatrix}\cos\theta_1 & -\sin\theta_1\\ \sin\theta_1 & \cos\theta_1\end{bmatrix}\begin{pmatrix}\sqrt{P_{x0}}\\ 0\end{pmatrix}. \quad (3)$$

If we consider that all recombination phases φ may be reached with equal probability when the wavelength varies, the mean PER is actually equal to $\langle\mathrm{PER}\rangle_\varphi$:

$$\langle\mathrm{PER}\rangle_\varphi = \frac{1}{2\pi}\int \frac{\cos^2\theta_1\cos^2\theta_2 + \sin^2\theta_1\sin^2\theta_2 - 2\cos\theta_1\cos\theta_2\sin\theta_1\sin\theta_2\cos\varphi}{\cos^2\theta_1\sin^2\theta_2 + \sin^2\theta_1\cos^2\theta_2 - 2\cos\theta_1\cos\theta_2\sin\theta_1\sin\theta_2\cos\varphi}\,\mathrm{d}\varphi. \quad (4)$$

The measured quantity with a white source would be equal to $\mathrm{PER}_{\mathrm{Int}}$ and not $\langle\mathrm{PER}\rangle_\varphi$ since

$$\frac{\int(\cos^2\theta_1\cos^2\theta_2 + \sin^2\theta_1\sin^2\theta_2 - 2\cos\theta_1\cos\theta_2\sin\theta_1\sin\theta_2\cos\varphi)\mathrm{d}\varphi}{\int(\cos^2\theta_1\sin^2\theta_2 + \sin^2\theta_1\cos^2\theta_2 - 2\cos\theta_1\cos\theta_2\sin\theta_1\sin\theta_2\cos\varphi)\mathrm{d}\varphi} = \mathrm{PER}_{\mathrm{Int}}. \quad (5)$$

In the following discussion we first explain in detail why so simple a device such as one PM patchcord exhibits a wavelength- and time-dependent PER (Section 2). Because PER measurement with a white source is forbidden, we consider a monochromatic source. We then explain why the usual scheme for PER measurement, i.e., the rotating polarizer, is most of the time pessimistic (Section 3). We show how to measure the PER of a connection composed of two connectors and one mating sleeve (Section 4). Finally, results are reported and the influence of the mating sleeve in PER is highlighted (Section 5). To conclude, we explain where a white source may be used (Section 6).

Unless otherwise specified, we assume that the signal travels along the PMF slow axis.

2. Frequency Evolution of the Polarization Extinction Ratio

The PER of a patchcord concatenation is mainly degraded by connectors (PER between 20 to 25 dB) and not by the fiber[7,12,13] (the PER is usually well above 40 dB). Therefore, if we neglect fiber contribution, a patchcord may be modeled by two wavelength-independent rotations, a wavelength-independent DGD and a wavelength-independent recombination phase φ (Fig. 1). The latter corresponds to the interferometric state for the central wavelength, which is assumed to be equal to 1053 nm. φ is equal to the total birefringence of the fiber modulo 2π. The total birefringence is of the order of $500 \times 2\pi$ (500 beat lengths) per meter of the PMF as we have measured a DGD equal to ~ 1.6 ps/m. This birefringence slightly changes with time because of environmental conditions (e.g., temperature, bending, pressure). Hence φ is stochastic from patchcord to patchcord and time varying. The transfer matrix of the patchcord is given by

$$M(f) = \begin{bmatrix}\cos\rho_2 & -\sin\rho_2\\ \sin\rho_2 & \cos\rho_2\end{bmatrix}\begin{bmatrix}\exp[+i(\pi f\mathrm{DGD} + \varphi/2)] & 0\\ 0 & \exp[-i(\pi f\mathrm{DGD} + \varphi/2)]\end{bmatrix}\begin{bmatrix}\cos\rho_1 & -\sin\rho_1\\ \sin\rho_1 & \cos\rho_1\end{bmatrix}, \quad (6)$$

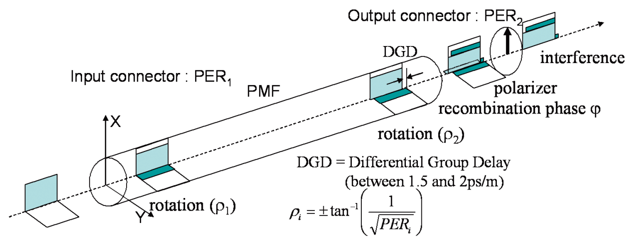

Fig. 1. (Color online) Pulse propagation in a patchcord composed of two connectors and a PMF.

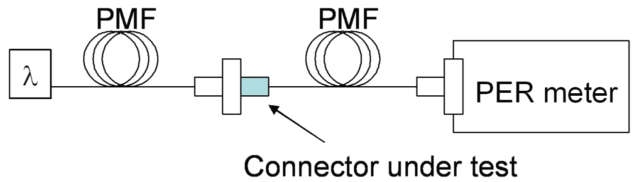

Fig. 3. (Color online) Connector PER measurement setup with a monochromatic source and a rotating polarizer.

where f is the frequency and the rotation angles ρ_i are linked to the connector PER by[7]

$$\mathrm{PER}_i = \frac{1}{\tan^2(\rho_i)}. \tag{7}$$

The PER of the patchcord is thus given by $P_x(f)/P_y(f)$, where $P_x(f)$ and $P_y(f)$ are calculated by

$$\begin{Bmatrix} \sqrt{P_x(f)}\exp[i\varphi_x(f)] \\ \sqrt{P_y(f)}\exp[i\varphi_y(f)] \end{Bmatrix} = M(f)\begin{pmatrix} \sqrt{P_{x0}} \\ 0 \end{pmatrix}. \tag{8}$$

One finds

$$\mathrm{PER} = \frac{\cos^2\rho_1\cos^2\rho_2 + \sin^2\rho_1\sin^2\rho_2 - 2\cos\rho_1\cos\rho_1\sin\rho_1\sin\rho_2\cos(2\pi f\mathrm{DGD}+\varphi)}{\cos^2\rho_1\sin^2\rho_2 + \sin^2\rho_1\cos^2\rho_2 + 2\cos\rho_1\cos\rho_1\sin\rho_1\sin\rho_2\cos(2\pi f\mathrm{DGD}+\varphi)}. \tag{9}$$

When ρ_1 and ρ_2 are small, which, it is hoped, is the case because PER_1 and PER_2 are usually high [according to Eq. (7), $\mathrm{PER}_i > 10$ dB, meaning that $\rho_i < 17.5°$], Eq. (9) may be approximated by

$$\frac{1}{\mathrm{PER}} \approx \frac{1}{\mathrm{PER}_1} + \frac{1}{\mathrm{PER}_2} + \frac{2}{\sqrt{\mathrm{PER}_1\mathrm{PER}_2}} \times \cos(2\pi f\mathrm{DGD}+\varphi). \tag{10}$$

Figure 2 gives a typical example of the variations of PER versus frequency. Such a curve has been experimentally measured in Ref. 7. It may be surprising that the patchcord PER depends on frequency whereas none of its elements depends on it, but a constant DGD means a linear variation of the phase versus frequency. Another way to understand the behavior of a patchcord is to consider it as any two-wave interferometer. As a matter of fact, as can be seen in Fig. 1, if a signal is launched on the X axis, because of the rotation of the input connector a small ghost is created on the Y axis and propagates at a different speed. At the output, the signal and its ghost are rotated again by the second connector. When looking at the X axis, one sees that the signal and the ghost of its ghost interfere. As in a Michelson or a Mach–Zehnder interferometer, two delayed waves interfere and make the output power vary with frequency. Because of energy conservation, when the power is not on the X axis, it is on the Y axis, making the PER a frequency function.

With such a frequency variation, it is not possible to measure the PER with a white source. As discussed in the introduction (Section 1), the measured quantity with a white source $\mathrm{PER}_{\mathrm{Int}}$ is not even equal to the average PER over wavelength. We now consider measurement with a monochromatic source and explain at the end in which cases a white source should still be used.

Note that, for a connector-free fiber, the geometrical reference is the X axis, either the slow or the fast axis. However, with connectors, the references are the connector keys, which are the only common references to different patchcords or to a patchcord and the lab.

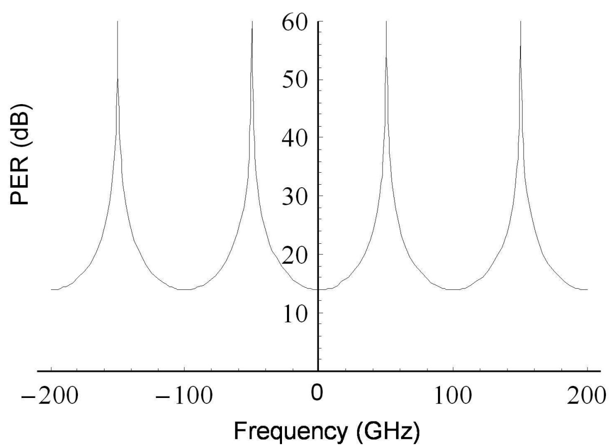

Fig. 2. PER of a patchcord composed of a PMF having 10-ps DGD (~7 m long) with connectors having a frequency-independent 20-dB PER.

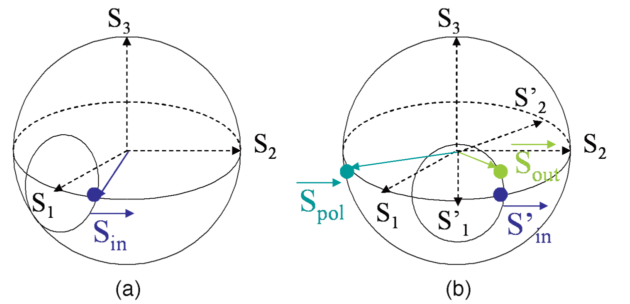

Fig. 4. (Color online) Poincaré spheres showing (a) the input SOP $\mathbf{S}_{in}$ of the fiber in the frame of the fiber axes and (b) the output SOP $\mathbf{S}_{out}$ in the frame determined by the output connector key in the case in which the connectors induce only a geometrical rotation of the axes (theoretical case).

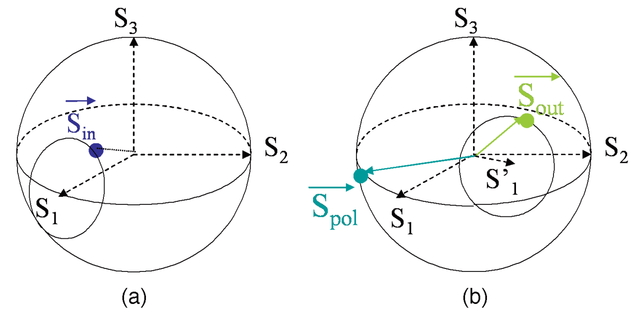

Fig. 5. (Color online) Same as Fig. 4 but in the case in which the connectors induce a full polarization transformation because of a twisting constraint (practical case).

3. Measurements with a Rotating Polarizer

Connector PER is usually measured with a commercial "PER meter" that consists of a rotating linear polarizer[1–3,13] (Fig. 3). However, the PER at the connector input is not infinite. Hence one usually replaces the input fiber with a free-space injection scheme, but angular accuracy between the free-space polarizer and the connector key is critical. Unfortunately, in practice a rotating polarizer measures only the polarization linearity ratio (PLR) of a monochromatic wave equal to P_{max}/P_{min}, where maximum and minimum are defined over the rotation of the polarizer. Confusion occurs because, most of the time, the acronym PER is also used in place of PLR. A mechanical or thermal stress of the patchcord under test changes the phase difference between the principal axes of the PMF. The minimum PLR during stressing is supposed to be equal to the PER. However, as we will show, this is valid only if the connector constraints on the fiber consist only of rotations of the axes. Figure 4(a) shows the Stokes vector[14] of the SOP of the monochromatic wave in the fiber in the frame of the fiber axes ($\mathbf{S}_1$ is the X axis). Because of the patchcord input fiber, the input SOP $\mathbf{S}_{in}$ is somewhere on the equator of the Poincaré sphere. Fiber birefringence rotates $\mathbf{S}_{in}$ over a circle centered on $\mathbf{S}_1$. Stress changes its position on the circle. Note that the circle radius gives the PER. Figure 4(b) shows the SOP in the frame of the output connector key. Because of the rotation of the output connector, the whole circle is rotated along the equator. The power at the output of the polarizer is given by $(1 + \mathbf{S}_{out}\,\mathbf{S}_{pol})/2$, where $\mathbf{S}_{pol}$ is the Stokes vector of the polarizer.[14] The minimum PLR, the ratio between the maximum and the minimum values of the output power, is obtained when $\mathbf{S}_{out}$ is farthest away from the equator. It can easily be shown that, in that case, PLR = PER.

However, connectors usually induce twisting constraints on the fiber. Connector rotation is not due to misalignment. If it were, an alignment accuracy of better than 0.5° (which is feasible and claimed by some companies) would yield much better PERs according to Eq. (7) than the one experimentally measured (23 dB specified by the same companies). Hence PER degradation is also due to the constraints applied by the connector on the fiber. These constraints may introduce a polarization transformation (azimuth and ellipticity) and not just a simple rotation (azimuth only). As a matter of fact, most of the time it is impossible to perfectly launch a signal on one of the PMF axes with a simple linear polarizer (see also measurement results in Section 5). Hence the ellipticity of $\mathbf{S}_{in}$ in the fiber axes' frame is not null in the general case. Figure 4 becomes Fig. 5: Neither $\mathbf{S}_{in}$ remains on the equator [Fig. 5(a)], nor does the circle remain centered on the equator [Fig. 5(b)]. With the same reasoning, when $\mathbf{S}_{out}$ is farthest away from the equator, its position is related not only to the circle radius but also to the ellipticity of the output connector. Hence measurements with a rotating polarizer are pessimistic. This result explains the measurements obtained in Ref. 3, in which discrepancies between two different measurement methods were reported and remained unexplained. One needs to directly measure the circle. This can be done with a polarimeter,[15] as we shall see in the next section.

4. Measurements with a Polarimeter

Basically a polarimeter may incorporate a rotating polarizer, but the rotation angle is accurately controlled and more processing is required. Other technologies also exist. The measurement setup is shown in Fig. 6. The polarization controller allows an accurate launch of the monochromatic wave on the slow axis of PM_A. As discussed in Section 3, a simple linear polarizer is not sufficient. Neglecting the PER of PM_A, we adjust the SOP at P in Fig. 6 by minimizing the size of the other circle that appears when stress is applied on PM_A. In practice PER at P is above 45 dB. Thus connection PERs (D_1 in Fig. 6) are measured instead of connector PERs. The full circle on the Poin-

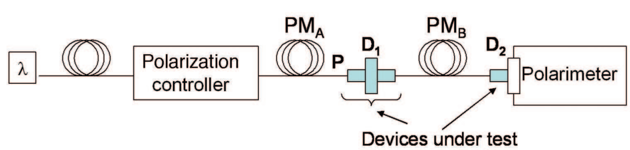

Fig. 6. (Color online) Connector PER measurement setup with a monochromatic source and a polarimeter.

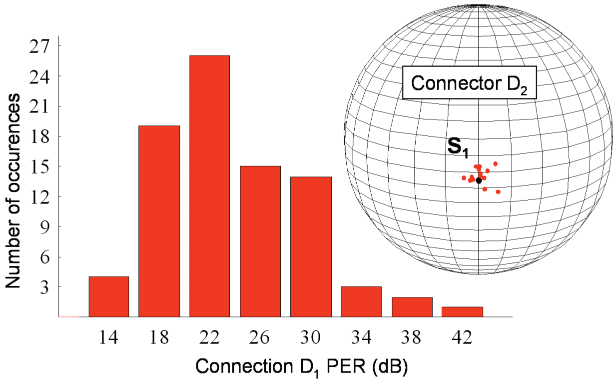

Fig. 7. (Color online) Histogram of connection D_1 PER and Poincaré sphere showing the output Stokes vector in the frame defined by the output connector D_2 key.

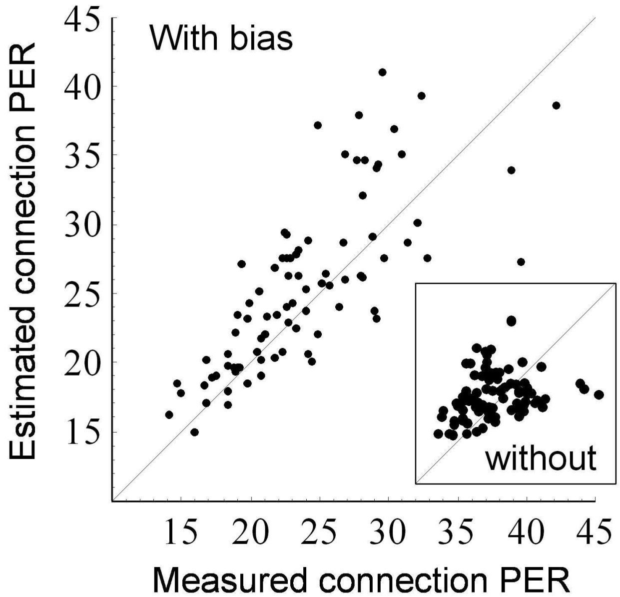

Fig. 8. Estimated connection D_1 PER with D_2 positions shown in Fig. 7 versus the actual measured values of the same connections D_1. The correlation factor is equal to 76%. This measurement bias used to plot this figure is $\theta_{\text{bias}} = -0.9°$, $\varepsilon_{\text{bias}} = +1.5°$. The inset is the same plot on the same scale without measurement bias. The correlation factor is only 27%.

caré sphere is measured when stress is applied to PM_B. The circle radius on the spheres in Fig. 5 still gives the PER of the connection D_1 between PM_A and PM_B (composed of two connectors and one mating sleeve). Provided that the polarimeter is sufficiently accurate, the circle center in Fig. 5(b) should also give the PER of the output connector D_2 of PM_B.

5. Measurement Results

We have developed an algorithm to determine both the circle radius and center. For a set of N patchcords, $2N$ connectors D_2 can be measured and $N \times (N - 2)/2$ different connections D_1 may be done because both connectors of the same patchcord cannot be connected together. This technique was applied to a set of seven FC/APC (fiber connector/angled physical contact) patchcords coming from the same supplier. All 14 connectors, D_2, and all 84 connections, D_1, were tested. We checked that our measurements depend on neither the choice of the axis (fast or slow) nor on the propagation direction of light. Measurement accuracy is limited by the PER of PM_A (usually > 45 dB, depending on its length[13]).

The PER of the connections D_1 and "position" of connectors D_2 on the Poincaré sphere are shown in Fig. 7. The position of a connector actually represents the SOP azimuth θ and ellipticity ε at the outside of the connector, provided the signal is on the slow axis of the fiber (X axis, $\mathbf{S}_1$) at its fiber side. Note that the ellipticity of D_2 is not constant in Fig. 7, proving that the connector induces not only a frame rotation but a full polarization transformation.

The PER of the D_2 connectors is given by

$$\mathrm{PER} = \frac{1}{\tan^2\rho} = \frac{1+\cos(2\rho)}{1-\cos(2\rho)} = \frac{1+\cos(2\theta)\cos(2\varepsilon)}{1-\cos(2\theta)\cos(2\varepsilon)}, \tag{11}$$

where ρ is the angle between the D_2 position and $\mathbf{S}_1'$.

From the position of connectors D_2, it is theoretically possible to determine the PER of any connection involving two connectors, provided that the influence of the mating sleeve is negligible. Consider two connectors, D_{2A} and D_{2B}, with respective position $\mathbf{S}_A$ defined by $(\theta_A, \varepsilon_A)$ and $\mathbf{S}_B$ defined by $(\theta_B, \varepsilon_B)$. When D_{2A} and D_{2B} are connected, they face each other as if they were in a mirror. Hence the portion of energy that is on $\mathbf{S}_A$ at the output of D_{2A} and that will be transferred to the slow axis of the fiber connected to D_{2B} is given by $(1 + \mathbf{S}_A \cdot \mathbf{S}_{-B})/2$ where $\mathbf{S}_{-B}$ is the Stokes vector defined by $(-\theta_B, -\varepsilon_B)$ to take the mirror effect into account. The PER of the connection is given by

$$\mathrm{PER}_{\mathrm{dB}} = 10\log_{10}\left(\frac{1+\vec{S}_A\cdot\vec{S}_{-B}}{1-\vec{S}_A\cdot\vec{S}_{-B}}\right). \tag{12}$$

However, this computation strongly depends on the polarimeter accuracy. The polarimeter manufacturer considers that the repeatability is ~0.1° (we measured a standard deviation of 0.2° with our connectors) and that the measurement bias is of the order of 1°. The bias may be inserted into Eq. (12) by a change of frame (for a small bias, changing θ into $\theta - \theta_{\text{bias}}$ and ε into $\varepsilon - \varepsilon_{\text{bias}}$ is almost valid). We have tried to determine the bias by maximizing the correlation factor between the measured connection D_1 PER and the theoretical values obtained with Eq. (12) and connector D_2 measurements. Figure 8 shows the estimated connection PER versus the measured connection PER. The optimum bias was found to be $\theta_{\text{bias}} = -0.9°$, $\varepsilon_{\text{bias}} = +1.5°$. The correlation factor is equal to 76% as compared with 27% without biasing the measurements. We assume that the correlation factor is still somewhat low,

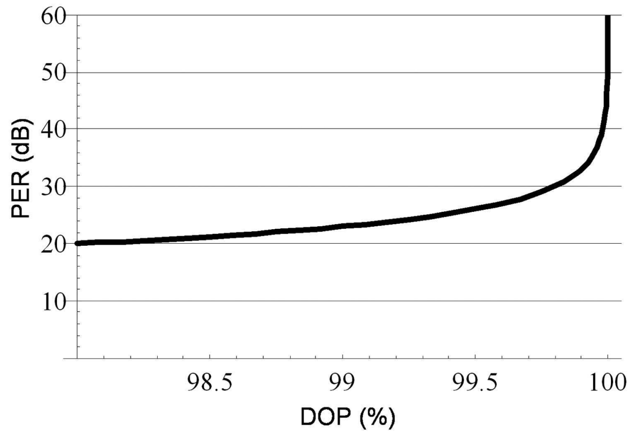

Fig. 9. PER of connection D_1 versus DOP when measurement is performed with a white source and the setup of Fig. 6.

mostly because of the constraints coming from the mating sleeve. Indeed, the connection PER depends on the way the FC/APC connectors are screwed to the mating sleeve.

6. Validity of Measurement with a White Source

As already stated, measurement with a polarized white source requires the PER to be frequency independent. This is the first issue.

When polarization transformers, such as connectors, at the device ends are present, the only valid measurement scheme is the polarization analyzer (Fig. 4). The measurement of the connection D_1 PER could theoretically be done with a white source. As a matter of fact, the different wavelengths comprising the white source will be located on a point of the circle in Fig. 5(b). All points of the circle are uniformly reached if the white-source bandwidth is much larger than the inverse of the PM_B fiber DGD. In that case, the average SOP at the patchcord output is $\mathbf{S}_1'$ in Fig. 5(b), which gives the D_2 PER, and the degree of polarization (DOP) is related to the circle radius and thus to the D_1 PER. It can be shown that[14]

$$\mathrm{PER} = \frac{1 + \mathrm{DOP}}{1 - \mathrm{DOP}}. \tag{13}$$

However, in practice, this scheme is not accurate because a small variation in the DOP induces an important variation in the PER, as can be seen in Fig. 9.

When there is no polarization transformers at the device end, as for a single PMF without connectors, the measurement can be performed with a rotating polarizer (analyzer). Moreover, with a white source the alignment of the input SOP to the slow axis of the fiber is easy[13] because the DOP is maximum when the alignment is perfect. The DOP is maximized by minimization of the optical power at the output of the rotating polarizer. This method with a white source and a rotating polarizer has been standardized for a PER PMF measurement.[16] Unfortunately it has been also extended to devices, like patchcords, that exhibit high PER wavelength dependence and full polarization transformers at their ends. Although such PER measurements would not be accurate in that case and could not be used for signal propagation modeling, they may be considered as good figures of merit as the $\mathrm{PER}_{\mathrm{Int}}$ is often degrading when $\langle \mathrm{PER} \rangle_\varphi$ is also degrading.

7. Conclusions

We have shown that, because of the high birefringence of the PMF, even a simple patchcord exhibits strong PER wavelength variations. This forbids the use of a white source for PER measurement of most devices. We have also shown that the classical method consisting of optical power measurements through a rotating polarizer is most often pessimistic. We have reported PER measurements with a polarimeter of connections (composed of two connectors and one mating sleeve) and connectors. The influence of the mating sleeve was demonstrated. Finally, we have defined the validity domain of the standardized PER measurement method by using a white source and a rotating polarizer.

References

1. N. R. Haigh and S. Kemsley, "Aspect of PER measurements," presented at the Optical Fiber Measurement Conference, Cambridge, UK, 26–28 September 2001; http://www.npl.co.uk/photonics/news/polarisation.html.
2. R. Stevens, "Polarisation extinction ratio—measurement requirements for optical communication systems," NPL Rep. CETM 41 (National Physical Laboratory, 2002).
3. R. M. Craig, "Interlaboratory comparison of polarization crosstalk measurement methods in terminated high-birefringence optical fiber," in *Optical Fiber Communication Conference (OFC)*, Vol. 2 of 1998 OSA Technical Digest Series (Optical Society of America, 1998), pp. 180–181.
4. M. K. Davis, A. Echavarria, and D. A. S. Loeber, "Polarization extinction ratio impact on spectral stability of Bragg grating stabilized laser diodes," IEEE Photon. Technol. Lett. **16,** 2003–2005 (2004).
5. G. Cancellieri, P. Fantini, and U. Pesciarelli, "Effects of joints on single-mode single-polarization optical fiber links," Appl. Opt. **24,** 964–969 (1985).
6. G. Cancellieri, P. Fantini, and M. Tilio, "Single-mode single-polarization fibers: Effects of residual polarization coupling," J. Opt. Soc. Am. A, **2,** 1885–1890 (1985).
7. J. E. Rothenberg, D. F. Browning, and R. B. Wilcox, "Issue of FM to AM conversion on the National Ignition Facility," in *Proceedings of the Third International Conference on Solid State Lasers for Application to Inertial Confinement Fusion*, W. H. Lowdermilk, ed., Proc. SPIE **3492,** 51–61 (1999).
8. M. Monnerie, "Polarization-maintaining single-mode fiber cables: Influence of joins," Appl. Opt. **20,** 2400–2406 (1981).
9. www.lmj-cea.fr and www.llnl.gov/nif/project
10. W. Zheng, "Automated fusion-splicing of polarization maintaining fibers," J. Lightwave Technol. **15,** 125–134 (1997).
11. S. L. A. Carrara, "Birefringent-fiber splice alignment," in *Fiber Optic Sensors IV*, R. T. Kersten, ed., Proc. SPIE **1267,** 24–28 (1990).
12. Y.-Z. Dai, "Relation between external stresses and the degradation of extinction ratio of polarization maintaining fibers,"

presented at the National Fiber Optic Engineers Conference, Denver, Col., 28–30 August 2000.
13. F. M. Sears, "Polarization-maintenance limits in polarization-maintaining fibers and measurements," J. Lightwave Technol. **8,** 684–690 (1990).
14. R. M. Azzam and N. M. Bashara, *Ellipsometry and Polarized Light* (Elsevier, 1989).
15. "FOTP-199: In-line polarization-crosstalk measurement method for polarization-maintaining optical fibers, components, and systems," Rep. TIA-455-199 (Telecommunications Industry Association, 2002).
16. "FOTP-193: Polarization crosstalk method for polarization maintaining optical fiber and components," Rep. TIA-455-193 (Telecommunications Industry Association, 1999).